BT301 - FINALTERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS

GROUP: BIOTECH BRAINY BUNCH

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Current finals – 2022

1- Define domain?

Large polypeptide chains usually fold into two or more globular clusters known as domains, which often give these proteins a bi- or multilobal appearance.

2- Multiplex PCR and Rt-PCR?

Multiplex PCR

Multiplex PCR is used to amplify several different DNA sequences simultaneously.

RT-PCR

Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase. Various strategies can be adopted for first-strand cDNA synthesis. Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase.

3- Define Ortholog?

Orthologs are genes in different species that evolved from a common ancestral gene by speciation, and, in general, orthologs retain the same function during the course of evolution.

Identification of orthologs is a critical process for reliable prediction of gene function in newly sequenced genomes.

4- Comparative genomics of eukaryotes?

The minimal eukaryotic genome is smaller than many bacterial genomes. Comparative genomics can be used to identify genes and regulatory elements. It gives insight into the evolution of key proteins.

5- Instability of plasmid?

Structural instability

Structural instability of plasmids may arise by deletions or rearrangements of DNA.

6- Central dogma of life?

Central dogma. The central dogma of molecular biology is a theory stating that genetic information flows only in one direction, from DNA, to RNA, to protein, or RNA directly to protein.

7- Significance of inverse PCR?

Inverse PCR

Inverse PCR is used to identify the flanking sequences around genomic inserts.

8- Role of PCR on environment?

PCR has widespread applications in various fields of life sciences including genetic engineering, Medical, Forensic, agriculture, environment. PCR has been used in cloning and screening of genomic libraries.

9- Role of RLFP

RFLP

RFLP-Restriction fragment length polymorphism exploits variations in homologous DNA sequences. In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis DNA fingerprinting.

10- Sequence of DNA?

DNA sequencing refers to the general laboratory technique for determining the exact sequence of nucleotides, or bases, in a DNA molecule. The sequence of the bases (often referred to by the

first letters of their chemical names: A, T, C, and G) encodes the biological information that cells use to develop and operate

11- Application of PCR?

Application of PCR

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc. PCR has been used in gene cloning and screening of genomic libraries

PCR-Medicine

PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis. Molecular tools have also allowed to perform prenatal genetic diagnosis

PCR-Forensic sciences

Forensic science is the application of scientific procedures to solve criminal and legal matters. Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

PCR-DNA profiling

DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

PCR-Agricultural sciences and environment

PCR has also facilitated research in detection of pathogens in plants, animals and environment. PCR has been used in cloning and screening of genomic libraries.

12- Chromosome jumping process?

Chromosome jumping

In chromosome jumping, the DNA of interest is identified, cut into fragments with restriction enzymes and circularized. It brings together DNA sequence that were originally located a considerable distance apart in the genome. These cloned DNAs from the closure sites make up a jumping library. One of the application of chromosome walking and jumping is in the cloning of human cystic fibrosis gene.

13- Primer?

PCR primers are short pieces of single-stranded DNA, usually around 20 nucleotides in length. Two primers are used in each PCR reaction, and they are designed so that they flank the target region (region that should be copied).

14- Primer extension?

Primer extension

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The single primer method. It involves priming in vitro DNA synthesis with a chemically synthesized oligonucleotide (7-20 bp) that carries a base mismatch with the complementary sequence. After E. coli transformation, heteroduplex that are formed are either that of original wild type DNA or that containing the mutated base.

15- Primary structure of protein?

Primary structure of proteins

Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain. Each type of protein has a unique amino acid sequence. Linus Pauling and Robert Corey carefully analysed the peptide bond. Their findings laid the foundation for our present understanding of protein structure. They demonstrated that the peptide C - N bond is somewhat shorter than the C – N bond in a simple amine. The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other. trans to each other. Pauling and Corey concluded that the peptide C - N bonds are unable to rotate freely because of their partial double-bond character. Rotation is permitted about the N - C and the C - C bonds. The bond angles resulting from rotations at C are labelled (phi) for the N – C bond and (psi) for the C - C bond. In principle, and can have any value between +180 & -180.

16- E.coli as plasmid?

Escherichia coli, perhaps the most-studied microorganism, has been found to possess a variety of plasmid types including those associated with virulence (91). Several types of E. coli virulence plasmids exist, including those essential for the virulence of enter toxigenic.

17- Database of DNA sequence?

DNA sequence databases

Since the development of current DNA sequencing technology large amount of sequence data has accumulated that is maintained in 3 data bases

i). National centre for Biotechnology Information

ii). DNA Databank of Japan

iii). European Bioinformatics institute-UK

18- Four cloning steps?

Steps in cloning with ʎ

Cloning with lambda replacement vectors involves the following steps

(i). Isolation and cutting vector DNA with appropriate restriction enzymes

(ii). Connecting the two lambda fragments to foreign DNA by using DNA ligase

(iii). In vitro packaging of recombinant DNA

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(iv). Infection of E. coli cells and isolation of phage clones by picking plaques on a host strain

(v). Checking recombinant phage for the presence of foreign DNA

19- High capacity vectors?

A YAC is a vector used to clone DNA fragments larger than 100 kb and up to 3,000 kb. YACs are useful for the physical mapping of complex genomes and for the cloning of large genes.

20- cDNA in cloning?

Complementary DNA (cDNA) cloning is termed for the gene cloning (cloning of DNA fragments) obtained from cDNA. The principle of cDNA cloning is that it involves the copying of mRNA transcripts into DNA, which are then inserted into bacterial plasmids and then placed into bacteria by transformation.

21- Role of Retrotransposons in human genome?

Retrotransposons constitute over 40% of the human genome and consist of several millions of family members. They play important roles in shaping the structure/evolution of the genome.

22- Orthologs in comparative genomics?

Orthologs are homologous genes in different organisms that encode proteins with the same function and which have evolved by direct vertical descent.

23- Principle strategy used in siRNA technology?

Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short-term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation.

24- How functional complementation restores the lost function in higher organism like animals?

Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype. Functional complementation is also possible in transgenic animals and plants. It has been used for complementation in transgenic mice to isolate the Shaker-2 gene.

25- Multiplex and Real time PCR?

Multiplex PCR Multiplex PCR is used to amplify several different DNA sequences simultaneously.

Real-time quantitative PCR (qPCR) qPCR is used to measure the quantity of a target sequence. It is used to measure starting amounts of DNA, cDNA, or RNA.

26- All the structures of protein?

Primary structure

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Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain. Each type of protein has a unique amino acid sequence. Proteins are different from one another on the basis of their amino acid sequence. Peptide Bond Is Rigid and Planar Linus Pauling and Robert Corey carefully analysed the peptide bond. Their findings laid the foundation for our present understanding of protein structure. They demonstrated that the peptide C - N bond is somewhat shorter than the C - N bond in a simple amine. The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other.

Secondary structure

Secondary structure of proteins refers to the local conformation of some part of a polypeptide. The most a few types of secondary structures are particularly stable and occur widely in proteins. Prominent are:-

1. -Helix A polypeptide chain is helically coiled around an imaginary axis and it makes -helix. It is a prominent structure in the proteins.

2. β- conformations Polypeptide chains stretch side by side longitudinally and form sheets. These sheets are called β – sheets.

Tertiary structure

The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's tertiary structure. It includes longer-range aspects of amino acid sequence. Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein. Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments. Large polypeptide chains usually fold into two or more globular clusters known as domains, which often give these proteins a bi- or multilobal appearance.

Quaternary Structure

Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different. The spatial arrangement of these subunits is known as a protein's quaternary structure. A multi-subunit protein is also referred to as a multimer. A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a promoter.

27- How copy number can be controlled by the effect of RNA-1?

The effect of plasmid copy number:

The number of ribosomes in a cell far exceeds any one class of mRNA. One way of increasing the expression of a cloned gene is to increase the number of the corresponding transcript.

28- Major technique use for genome characterization?

Major Techniques used for Genomes Characterization

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- Cloning
- Hybridization
- PCR amplification
- Sequencing
- Computational tool
- Genomes digested with restriction enzymes and inserted in vectors to produce genomic libraries. BACs
- YACs
- Genomes Characterization Techniques – Hybridization. To arrange large contigs of genomes to produce genetic maps and physical maps of genomes. To arrange large contigs of genomes to produce genetic and physical maps.
- Genomes Characterization Techniques – PCR. Technique to amplify the DNA. Different variants of the technique used
- Genomes Characterization Techniques – DNA Sequencing. One of the important technique used to characterize the genomes. To study structure and function of genomes. DNAs are amplified and sequenced
- Genomes Characterization Techniques – Computational Tools. Used to align the sequenced DNA to produce physical maps of the genomes.

29- Reverse transcriptase?

Reverse transcriptase (RT), also known as RNA-dependent DNA polymerase, is a DNA polymerase enzyme that transcribes single-stranded RNA into DNA. This enzyme is able to synthesize a double helix DNA once the RNA has been reverse transcribed in a first step into a single-strand DNA.

30- Benefits of genomic research?

Benefits of Genomes Research

Genomes Research – Molecular Medicine

Improve diagnosis of disease. Detect genetic predispositions to disease (cancer, diabetes etc). Create drugs based on molecular information. Use gene therapy and control systems as drugs

Genomes Research – Risk Assessment

Evaluate the health risks faced by individuals who may be exposed to radiations and. to cancer causing chemicals and toxins. Genomes Research – Bioarchaeology, Anthropology, Evolution and Human Migration. Study evolution through genetic variants in lineages. Study of migration of different populations. Study mutations on the Y chromosome to trace lineage and migration of males. Evolution of mutations with ages of populations

Genomes Research – DNA Forensics

Identify potential suspects whose DNA may match evidence left at crime scenes. Exonerate persons wrongly accused of crimes. Identify catastrophe victims. Establish paternity and other

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family relationships. Identify endangered and protected species as an aid to wildlife officials. Genomes Research. Detect bacteria and other organisms that may pollute air, water, soil and food. Match organ donors with recipients in transplant programs. Determine pedigree for seed or livestock

Genomes Research – Disease-resistant crops and disease-resistant animals

Grow disease/insect resistant and drought-resistant crops. Breed healthier, more productive, disease-resistant farm animals. Develop bio pesticides. Incorporate edible vaccines incorporated into food products.

Genomes Research – Microbial Genomics.

Rapidly detect and treat pathogens (disease-causing microbes). Develop new energy sources (biofuels). Monitor environment to detect pollutants. Protect populations from biological and chemical warfare. Clean up toxic waste safely and efficiently

31- What is biotechnology? Write its field name?

Introduction to Biotechnology

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.

Traditional biotechnology refers to the conventional techniques that have been used for many centuries to produce beer, wine, cheese etc. Modern Biotechnology embraces all methods of genetic modification by recombinant DNA & cell fusion techniques together with the modern Developments of traditional biotechnological processes.

White Biotechnology

Development of processes and microorganisms for. Industrial processes. **Key concept** Scale of operation Low volume high price

Example: Enzyme production image1

Red Biotechnology

It is concerned with the discovery and development of innovative drugs and treatments

Green Biotechnology

It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones. **Example:** Bt corn from Syngenta, Bt cotton from Monsanto.

32- Oligonucleotide?

Oligonucleotides, or oligo, are short single strands of synthetic DNA or RNA that serve as the starting point for many molecular biology and synthetic biology applications! From genetic

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testing to forensic research and next-generation sequencing, an oligo may very well be the starting point.

33- Cosmid Vector?

Plasmids have been constructed which contain a fragment of ʎ DNA including 'cos' site. These plasmids have been termed cosmids and can be used as gene-cloning vectors in conjunction with the in vitro packaging system. Cosmids provide an efficient means of cloning large pieces of foreign DNA. Cosmids are particularly attractive vectors for constructing libraries of eukaryotic genomic fragments.

34- Bacteriophage?

Bacteriophages, also known as phages, are viruses that infect and replicate only in bacterial cells. They are ubiquitous in the environment and are recognized as the most abundant biological agent on earth. They are extremely diverse in size, morphology, and genomic organization.

35- Insulin as highly expressed?

What happens if I have too much insulin? If a person accidentally injects more insulin than required, e.g. because they expend more energy or eat less food than they anticipated, cells will take in too much glucose from the blood. This leads to abnormally low blood glucose levels (called hypoglycaemia).

36- Expression libraries in prokaryotes and yeast?

Yeast Library:

Yeast display libraries are constructed by cloning the genes coding for antibody fragments into vectors that can be transformed into the yeast cells.

Prokaryotes library

In prokaryotes, the genome is composed of a single, double-stranded DNA molecule in the form of a loop or circle. The region in the cell containing this genetic material is called a nucleoid. Some prokaryotes also have smaller loops of DNA called plasmids that are not essential for normal growth.

37- Cloning strategies?

Any cell based cloning procedure has four essential parts

i). A method of generating DNA fragment for cloning

ii). A reaction that inserts that fragment into the chosen cloning vectors

iii). A means for introducing that recombinant vector into a host cell wherein it is replicated

iv). A method of selecting recipient cells that have acquired the recombinant

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Cell based cloning strategy, is to divide the source DNA into manageable fragments and clone everything.

38- Spotted DNA arrays?

Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR product individually onto a solid support where they are immobilized.

39- DNA profiling?

It is the process where a specific DNA pattern, called a profile, is obtained from a person or sample of bodily tissue. Even though we are all unique, most of our DNA is actually identical to other people's DNA. However, specific regions vary highly between people. These regions are called polymorphic.

40- RAPD?

RAPD-Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random. RAPD-Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random.

41- Chain termination or deoxy procedure?

This method was developed by Sanger et al. (1977). It capitalizes on the ability of DNA polymerase to incorporate dideoxynucleoside triphosphates as substrate. Dideoxynucleoside triphosphate act as chain terminator. Chain termination or deoxy procedure DNA synthesis is carried out in the presence of the four deoxy nucleoside triphosphates, one or more of which is labelled with 32P and in four separate incubation mixes.

42- Strategy cloning?

A serious disadvantage of the hairpin method is that cleavage with S1 nuclease results in the loss of sequences at the 5’ end of the clone. This strategy has therefore been superseded with other methods. For cDNA expression libraries, it is advantageous if the cDNA can be inserted into the vector in the correct orientation. This can be achieved by self-priming method by adding linker molecule to double stranded cDNA.

43- RFLP significance?

RFLP-Restriction fragment length polymorphism exploits variations in homologous DNA sequences. In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis DNA fingerprinting.

44- Screening strategies?

Major screening strategies involve

1). Genetic methods

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2). Sequence-dependent screening

3). Screening expression libraries

All useful vector molecules carry a selectable genetic marker or property. Plasmid and cosmid vectors carry drug resistance or nutritional marker. In phage vectors, plaque formation is itself the selected property

45- Epitope in antigen?

The small site on an antigen to which a complementary antibody may specifically bind is called an epitope or antigenic determinant. This is usually one to six monosaccharides or five to eight amino acid residues on the surface of the antigen

Example:

Common epitopes used for this purpose are Myc-tag, HA-tag, FLAG-tag, GST-tag, 6xHis, V5-tag and OLLAS. Peptides can also be bound by proteins that form covalent bonds to the peptide, allowing irreversible immobilisation.

46- Knock in and Knock out?

A gene knock out is a genetic technique in which one of an organism's gene is made inoperative that is simply called as knocked out. In molecular cloning, a knock-in refers to a gene manipulation method that involves the insertion of a protein coding cDNA sequence at a particular locus in an organism chromosome.

47- STS in physical maps?

Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use southern blotting. STS in physical maps. Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use southern blotting.

48- Yeast artificial chromosomes (YACs)?

These vectors replicate in yeast like normal chromosome but they have sites where very large DNA fragments can be inserted. YACs with inserts more than 150,000 bp are nearly as stable as normal cellular chromosomes.

49- Bacterial Genomes?

Small organisms carry high coding density (85-90%). 1 gene per 1000 bases in prokaryotes. Large variation in genome size between bacteria. Genomes of Bacteria – Large Variation Tremblaya princeps 140kb, 121 coding sequences

Sorangium cellulosum

14000kb

11599 coding sequences

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50- Thermophlic enzyme used in PCR?

Taq DNA polymerase

Taq polymerase is a thermostable DNA polymerase isolated from thermophilic bacterium Thermus aquaticus and is used in PCR to amply DNA fragments.

51- PCR and Types?

Polymerase chain reaction (PCR)

The impact of PCR upon molecular biology has been profound. The reaction is easily performed and leads to the amplification of specific DNA by an enormous Factor.

Types of PCR

Recently, variants have been developed from basic PCR method to improve performance and specificity and to amplify other molecules such as RNA

RT-PCR

Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase. Various strategies can be adopted for first-strand cDNA synthesis

Competitor RT-PCR

One use of RT-PCR is in determining the amount of mRNA in a sample (competitor RT-PCR).

Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase

Real-time quantitative PCR (qPCR)

qPCR is used to measure the quantity of a target sequence. It is used to measure starting amounts of DNA, cDNA, or RNA

Nested PCR

In nested PCR, two sets of primers are used in two successive PCRs. First set of primers allows a first amplification. Product of first PCR is subjected to a second PCR using the second set of primers

Inverse PCR

Inverse PCR is used to identify the flanking sequences around genomic inserts

Multiplex PCR

Multiplex PCR is used to amplify several different DNA sequences simultaneously.

52- Benton and Davis' plaque lift procedure?

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Benton and Davis (1977) devised a method called plaque lift, in which nitrocellulose filter is applied to the upper surface of agar plates, making direct contact between plaques and filter. Number of alternative labelling methods are available that avoid the use of radioactivity. It included digoxigenin or biotin.

53- Genetic Drift?

Change in the frequency of a gene variant (allele) in a population due to random sampling of organisms.

Example

Humans Skin Colour

Past Finals subjective – 2021+ 2020

54- Difference between genetics and genomics?

Genetics

Genetics is the study of heredity, or how the characteristics of living organisms are transmitted from one generation to the next generation through DNA

Genomics

Genomics, in contrast, is the study of the entirety of an organism's genes - called the genome

55- What is size diff between LINES and SINES?

SINES (short interspersed elements) 200-300 bp • 100,000 copies

LINES (long interspersed elements) 1-5 kb • 10-10,000 copies

56- What is green biotechnology and give an example?

Green Biotechnology It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones.

Example Bt corn from Syngenta Bt cotton from Monsanto

57- What is Novel Diagnostic?

Complex systems biology, developmental genetics. To provide platform for microchips and DNA microarrays. Gene expression - RNA Complex systems biology, developmental genetics and genomics

58- Write only name of 4 Major technique used in genome characterization?

1) Cloning

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2) Hybridization

3) PCR Amplification

4) Sequencing

5) Computational tool

59- What is Insertional Inactivation?

Insertional inactivation is the inactivation of a gene upon insertion of another gene inside in its place or within its coding sequence

60- How many Standard Amino Acids are there? Write name of any 5?

There are 20 amino acids know as standard amino acids.

1) Leucine

2) Isoleucine

3) Valine

4) Proline

5) Methionine

61- What are various factors which affect mobility of DNA in the Gel?

Concentration of gel either high concentration or lower concentration. Ionic strength of buffer. Voltage Ethidium bromide (A chemical used as a stain to dye gel).

62- What is the role of Electroporation?

In Electroporation electric field is applied to cells in order to increase permeability of the cell membrane to take DNA

63- Define Autoradiography?

A technique using X-ray film to visualize fragments of molecules that have been radioactively labelled.

EX: It can be used to analyse the length and number of DNA fragments separated by gel electrophoresis

64- Write main sources and usage of citric acid?

Main source of citric acid is citrus fruits but in industrial production citric acid is obtained by Aspergillus Niger. It is used to preserve food and to flavouring food like vinegar

65- Define Gene?

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A gene is the basic physical and functional unit of heredity. Genes, which are made up of DNA, act as instructions to make molecules called proteins

66- Define Isopynic Centrifugation?

A method where the components of a sample (e.g. DNA) are separated on the basis of their density.

67- Define Autoradiography?

A technique using X-ray film to visualize fragments of molecules that have been radioactively labelled. For example, it can be used to analyse the length and number of DNA fragments separated by gel electrophoresis.

68- Define: Euchromatic & heterochromatin?

Highly condensed chromatin is known as heterochromatin. The more extended form is known as euchromatin.

69- What is the main components consisting the RNA molecule?

• Each ribonucleotide is composed of three components:

A ribose sugar, A Nitrogenous Base, A Phosphate group

70- What is .palindromic sequence?

A DNA locus whose 5'-to-3' sequence is identical on each DNA strand. The sequence is the same when one strand is read left to right and the other strand is read right to left. Recognition sites of many restriction enzymes are palindromic.

71- Define the function of Plasmid DNA ?

Plasmids are mostly circular DNA molecules that replicate separately from the host chromosome Plasmid help bacteria to survive stress

72- Ancient Pregnancy Test?

A papyrus described a test in which a woman who might be pregnant could urinate on wheat and barley seeds over the course of several days: “If the barley grows, it means a male child. If the wheat grows, it means a female child. If both do not grow, she will not bear at all.”

73- What is pseudo gene?

Pseudogenes are genes that have lost their function. They have lost their gene expression in the cell or their ability to code protein. The term was coined in 1977. Pseudogenes can result from mutations in a gene whose product is not needed for the survival of the organism

74- Differentiate between multimer, oligomer and promoter?

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A multi-subunit protein is also referred to as a multimer. A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a promoter. Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.

75- Write one differential point between super coiled and relaxed loops?

Genetic Material in Bacteria some of the loops are super-coiled while some are relaxed. Relaxed loops are formed as a result of a nick introduced into super-coiled loops by a cellular DNase. If a super-coiled DNA molecule receives a nick, the strain of under winding is immediately removed, and all the super-coiling is lost. The super-coiled loops are dynamic structures which change during cell growth & division. An E. coli chromosome is estimated to have about 400 supercoiled loops. The DNA compaction in a bacterial cell is contributed by supercoiling of loops, macromolecular crowding and DNA binding proteins.

76- What is Selfish DNA?

Selfish DNA is a term for sequences of DNA that have two distinct properties: the DNA sequence spreads by forming additional copies of itself within the genome; and, it makes no specific contribution to the reproductive success of its host organism. (It may or may not have significant negative effects.)

77- Side effect of ethidium bromide?

Ethidium bromide can bind with DNA, it is highly toxic as a mutagen. It may potentially cause carcinogenic or teratogenic effects, although no scientific evidence showing either health effect has been found. Exposure routes of ethidium bromide are inhalation, ingestion, and skin absorption. An acute exposure to ethidium bromide causes irritation of the mouth, upper respiratory tract, skin, and eyes.

78- Importance of origin region in plasmids?

The replicon is comprised of the origin of replication (ORI) and all of its control elements. The ORI is the place where DNA replication begins, enabling a plasmid to reproduce itself as it must to survive within cells. • When plasmids are present in microbes, whether they have narrow host range or wide host range. • Plasmids encode only a few of the proteins required for their own replication. Host range of plasmid is determined by ori region (Point where DNA replication mainly starts) Plasmids whose ori derived from Col E1 have restricted host range RP4 and RSF1010 and many plasmids from Staphylococcus aureus have broad host range

79- Write to most significant function of Nucleic Acid / DNA Function?

The main functions is store and transfer genetic information. To use the genetic information to direct the synthesis of new protein. The deoxyribonucleic acid is the storage for place for genetic information in the cell

80- Define Natural Plasmid?

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Plasmid which were not constructed in vitro for the sole purpose of cloning are called natural plasmids Example Col E1 is natural occurring plasmid.

81- Write 2 main factors that result in High yield of Plasmid DNA?

2 main factors that result in High yield of plasmid are

1) Concentration of chemical that is used Calcium chloride

2) Temperature

82- Write line for Influenza?

The influenza virion (as the infectious particle is called) is roughly spherical. It is an enveloped virus - that is, the outer layer is a lipid membrane which is taken from the host cell in which the virus multiplies

83- Explain Technique of Electroporation?

A rapid and simple technique for introducing cloned genes into a wide variety of microbial, plant and animal cell, including E. coli, is electroporation. In electroporation, electric field is applied to cells in order to increase permeability of the cell membrane to take DNA

Transformation with other organisms

E. coli often remains the host of choice for cloning Bacillus subtilis (preferable choice because it is natural competent) is a particularly attractive alternative cloning host. Animal cells, protoplasts (without cell wall) of yeast and plant are susceptible to transformation by liposomes. Other artificial method for transformation include gene gun. Lipofaction is a technique in which we use liposomes to transform protoplasts. Liposomes are vesicles made up of phospholipids and are positively charged. They are positively charged because mainly they are made from cationic lipids. They have the ability to enclose negatively charged DNA. After encapsulation liposomes fuse with the cytoplasmic membrane and then it can deliver DNA from the vesicle to the target cell. So lipofaction is also used to transform different type of host cells.

84- Write Nomenclature of Restriction Enzyme Modification?

Nomenclature

After the discovery of large number of Type II endonucleases, scientists proposed that there must be a naming mechanism which we can call as the nomenclature of the restriction endonucleases.

• A suitable system was proposed by Smith and Nathans (1973)

The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet . E. coli = Eco H. influenzae = Hin Strain identification is written as Ecok In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. influenzae Hindl, Hindil, HindlIl etc All restriction

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enzymes have general name endonuclease R and modification-methylase M followed by the system name, for example, in case of H. influenzae R. Hindill or M. Hindill

85- Explain R-M Phenomena?

During 1960s phage biologist elucidated the phenomenon of restriction and modification Host-controlled restriction/ modification. Restriction system allow bacteria to monitor the origin of incoming DNA. When the incoming DNA is a bacteriophage genome, the effect is to reduce the efficiency of plating (Efficiency of plating means the number of plaques that are produced during the lysis) i.e. to reduce the number of plaques in plating test The phenomenon of restriction and modification were well illustrated and studied by the

- E.coli behaviour of phage on two E. coli strains

EOP = 104 E. coli C and E. coli K Restrictive host must, of course protects its own DNA from restriction endonucleases by modification

Modification involves methylation of certain bases that constitute recognition sites E. coli K

86- Write Benefits of Genomic Research in Molecular Medicine Genomes Research - Molecular Medicine?

• Improve diagnosis of disease

• Detect genetic predispositions to disease (cancer, diabetes etc.)

Create drugs based on molecular information

• Use gene therapy and control systems as drugs Q. Chargaff Rule The work of Chargaff led him to following conclusions, also called “Chargaff Rules": 1. Base composition of DNA varies from one species to another. 2. The DNA isolated from different tissues of the same species has the same base composition. 3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment. 4. In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C). 5. It means that the sum of the purine residues equals the sum of the pyrimidine residues (AG=TC).

87- Biotechnology define and write its branches?

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.

White Biotechnology:

Development of processes and microorganisms for Industrial processes. e.g; Enzyme production

Red Biotechnology:

It is concerned with the discovery and development of innovative drugs and treatments.

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Green Biotechnology:

It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones. An additional gene was inserted into the normal genome of this corn that triggers the production of proteins that then protect the plant from harmful insects. This reduces the need for pesticides and increases crop yields.

Example:

Bt corn from Syngenta Bt cotton from Monsanto

88- Work of Franklin and Wilken's Work of Franklin & Wilkins (1950s)?

Rosalind Franklin and Maurice Wilkins performed the X-ray diffraction analysis of DNA fibres. They showed that DNA produces a characteristic X-ray diffraction pattern. From this pattern, they made two important findings: DNA molecules are helical. The helices have two periodicities along their long axis, a primary one of 3.4 Å and a secondary one of 34 Å.

89- Explain Structure of proteins ?

Primary structure of proteins:

Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain. Each type of protein has a unique amino acid sequence. Proteins are different from one another on the basis of their amino acid sequence. Secondary structure of proteins: Secondary structure of proteins refers to the local conformation of some part of a polypeptide. A few types of secondary structures are particularly stable and occur widely in proteins. The most prominent are: - 1. A-helix: - 2. B- conformations: Tertiary Structure the overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's tertiary structure. It includes longer-range aspects of amino acid sequence. Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein. Large polypeptide chains usually fold into two or more globular clusters known as domains, which often give these proteins a bi-or multilobal appearance. Quaternary Structure of Proteins. Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different. The spatial arrangement of these subunits is known as a protein's quaternary structure.

90- What is insertional inactivation?

Insertional inactivation is the inactivation of a gene upon insertion of another gene inside in its place or within its coding sequence. This video also explains the selection of recombinant colonies in selective antibiotic medium by insertional inactivation

91- What is DNA sequence; benefits and applications?

DNA sequencing is the process of determining the precise order of nucleotides within a DNA molecule. It includes any method or technology that is used to determine the order of the four bases—adenine, guanine, cytosine, and thymine—in a strand of DNA.

DNA sequencing: Benefits and Applications:-

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DNA sequence information is a prerequisite for planning any substantial manipulation of the DNA. Information is useful to various fields' i.e. molecular and evolutionary biology, metagenomics, medicine, forensics etc.

92- PCR & RT-QPCR and features?

PCR:

PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a Particular DNA sequence.

Real-time quantitative PCR (qPCR):-

RT-QPCR is used to measure the quantity of a target sequence. It is used to measure starting amounts of DNA, cDNA, or RNA

93- What is multiplex PCR?

Multiplex PCR

Multiplex PCR is used to amplify several different DNA sequences simultaneously

94- siRNA strategy?

Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short-term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation.

95- What is mean by Orthology?

Orthologs are homologous genes in different organisms that encode proteins with the same function and which have evolved by direct vertical descent.

96- What are relaxed plasmids?

These plasmids contain DNA regions called par-locuses (from partition) that control the correct division of the plasmids in the two daughter cells. Plasmids with high copy number are under relaxed control of replication. These plasmids are replicated independently of the replication of the bacterial chromosome

97- Applications of PCR?

Application of PCR

PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc. PCR-Gene cloning and expression. PCR has been used in gene cloning and screening of genomic libraries

PCR-Medicine

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PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis. Molecular tools have also allowed to perform prenatal genetic diagnosis

PCR-Forensic sciences

Forensic science is the application of scientific procedures to solve criminal and legal matters. Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

PCR-DNA profiling

DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

PCR-Agricultural sciences and environment

PCR has also facilitated research in detection of pathogens in plants, animals and environment

PCR-Molecular Palaeontology

Molecular palaeontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains

98- Write a note on plasmids and natural plasmids?

Plasmids

Plasmids are mostly circular DNA molecules that replicate separately from the host chromosome. Naturally occurring bacterial plasmids range in size from 5,000 to 400,000 bp.

Natural Plasmids:-

Natural plasmids as cloning vehicles. Plasmids which were not constructed in vitro for the sole purpose of cloning are called natural plasmids. Col E1 is a natural occurring plasmid. RSF2124 is a derivative of Col E1

99- 4 steps of cloning?

Steps of cloning:

Cloning with lambda replacement vectors involves the following steps:

(i). Isolation and cutting vector DNA with appropriate restriction enzymes

(ii). Connecting the two lambda fragments to foreign DNA by using DNA ligase

(iii). In vitro packaging of recombinant DNA

(iv). Infection of E. coli cells and isolation of phage clones by picking plaques on a host strain

(v). Checking recombinant phage for the presence of foreign DNA

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100- Explain Spotted DNA array?

Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR products individually onto a solid support where they are immobilized.

101- Write the uses of the DNA microarrays?

DNA Microarrays

Microarrays are used to study the expression of many genes at once. It involves the immobilization of thousands of gene sequences on glass slide that are then detected with complementary base pairing between the sample and the gene sequences on the chip.

102- Explain briefly Shotgun sequencing?

Shotgun sequencing is used for sequencing long DNA strands ▪ In this strategy, DNA shredded into smaller fragments that can be sequenced individually and then reassembled into original order.

103- What is a Sequencing genome?

The process of determining the exact order of nucleotides within a DNA or RNA molecule is called sequencing ▪ In 1970s, DNA sequencing was a very tedious process which involved determining only a few hundred nucleotides at a time

104- What do you know about chromosome jumping?

In chromosome jumping, the DNA of interest is identified, cut into fragments with restriction enzymes and circularized. It brings together DNA sequence that were originally located a considerable distance apart in the genome

105- Define functional complementation?

Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype

106- What do you know about immunochemical screening?

Immunochemical screening involves the use of antibodies that specifically recognize antigenic determinants on the polypeptides synthesized by a target clone

107- What are the requirement for expression in E.coli?

Synthesis of functional protein depends upon transcription of appropriate gene, efficient translation of mRNA and in many cases, posttranslational processing and compartmentalization of nascent polypeptide

108- .What is sequencing accuracy?

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Sequences that were read beyond 4oo bp contained an average of 3.2% error, while those less than 400 bp had 2.8% error

109- NAME three DATA base sequences use in DNA sequencing?

National centre for Biotechnology Information ii). DNA Databank of Japan

iii). European Bioinformatics institute-UK

110- Define mutagenesis?

Mutagenesis is a process to change the genetic information of an organism. It can occur naturally, or as a result of exposure to mutagens or induced experimentally in laboratory

111- What is a degenerate primer?

.A degenerate is actually a mixture of primers, all of similar sequence, but with variation with one or more positions.

112- Define RT-PCR?

RT-PCR(Reverse transcriptase) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by various strategies.

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